Introduction to Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software capable of performing tasks that normally require human intelligence. These tasks include understanding language, recognizing images, making decisions, solving problems, learning from experience, and interacting with users.

Common applications: virtual assistants, recommendation systems, healthcare diagnosis, fraud detection, autonomous vehicles, robotics, smart search, and personalized learning.

2. AI, Machine Learning and Deep Learning

Artificial Intelligence (AI) is the broad field of building intelligent systems. Machine Learning (ML) is a subset of AI in which systems learn patterns from data instead of relying only on explicitly written rules. Deep Learning is a subset of ML that uses neural networks with multiple layers to learn complex patterns from large amounts of data.

AI $\rightarrow$ Machine Learning $\rightarrow$ Deep Learning

3. Types of Machine Learning

• Supervised Learning: The model learns from labeled examples. Examples include spam detection, house-price prediction, and student-placement prediction.

• Unsupervised Learning: The model discovers patterns or groups in unlabeled data. Clustering customers into different groups is a common example.

• Reinforcement Learning: An agent learns by interacting with an environment and receiving rewards or penalties. It is used in robotics, games, and decision-making systems.

4. Neural Networks

A neural network is a machine learning model inspired by biological neurons. It contains interconnected nodes organized into layers. During training, the network adjusts its parameters to reduce prediction errors. Deep neural networks contain many layers and can learn highly complex representations.

Generative AI and Modern AI Systems

5. What is Generative AI?

Generative AI is a class of AI systems that can create new content based on patterns learned from training data. It can generate text, images, audio, video, software code, summaries, and other forms of content.

Examples: AI chatbots can generate text, image-generation models can create pictures from descriptions, coding assistants can generate programs, and speech models can generate or transform audio.

6. Large Language Models (LLMs)

Large Language Models (LLMs) are AI models trained on large collections of text and other data. They learn patterns in language and can perform tasks such as answering questions, summarizing documents, translating text, extracting information, and generating code. Examples of widely known LLM-based systems include GPT and Gemini. Users interact with these systems through prompts, which provide instructions or context for the desired output.

7. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of relying only on what the model learned during training, a RAG system first searches a document collection for relevant passages and then gives those passages to the LLM as context. This helps the model give answers grounded in specific documents and reduces made-up answers (hallucinations).

Embeddings, Vector Databases and Summary

8. Embeddings

An embedding is a list of numbers (a vector) that represents the meaning of a piece of text. Texts with similar meaning have vectors that are close to each other. Embeddings make semantic search possible, which means finding passages by meaning rather than by exact keywords.

9. Vector Databases

A vector database stores embeddings and quickly finds the ones most similar to a query vector. FAISS is a popular library for similarity search. Chroma and Pinecone are other well-known vector stores. In a RAG system, the user's question is converted into an embedding, and the vector database returns the top-k most similar text chunks.

10. Key Takeaway

AI is a broad field that includes Machine Learning and Deep Learning. Generative AI extends these capabilities by creating new content, while LLMs enable powerful language-based applications. Technologies such as embeddings, vector databases, and RAG make it possible to build AI systems that can work with private, domain-specific, and frequently changing information.